

PCAP Malware Investigation Report

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1 Introduction

This report reviews network traffic captured in a PCAP file to identify signs of malware infection and possible data exfiltration. Packet capture analysis helps investigators rebuild network activity and spot suspicious communication between internal hosts and external systems.

The goal of this investigation is to determine whether malicious behavior exists in the capture, examine how the malware may have been delivered, and understand how the infected system communicated after compromise.

2 Overview

The packet capture was examined using several common network forensic tools:

- Wireshark for detailed packet inspection
- Security Onion alerts for intrusion detection
- NetworkMiner for artifact extraction

Together, these tools help investigators trace suspicious traffic, recover transferred files, and identify communication between compromised systems and outside servers.

3 Malicious Executable Detection

The analysis revealed evidence of a suspicious executable being transferred across the network. One major indicator was the presence of an **MZ header**, which is a well-known signature for a Windows executable file.

This strongly suggests that a Windows-based executable was downloaded from an outside source. In a forensic context, that kind of activity is often linked to malware delivery through phishing or malicious downloads.

The MZ header is a strong indicator that a Windows executable entered the environment from an external source. That pattern is consistent with malware delivery behavior commonly seen in phishing and drive-by download cases.

4 NetworkMiner Host Identification

The PCAP was also reviewed in **NetworkMiner**, which helped identify the systems involved and extract relevant artifacts from the traffic.

NetworkMiner showed the host tied to the suspicious transfer and supported the conclusion that the executable moved from an external server to an internal machine. This matters because it helps identify which host was likely compromised and where the malware came from.

5 POST Commands and Data Exfiltration

Further packet analysis showed multiple **HTTP POST requests** originating from the infected host.

Those POST requests appeared to send information from the compromised machine to an outside server. The transmitted content included system-related details such as running processes. This behavior is common in malware cases because POST requests are frequently used for command-and-control traffic and data exfiltration.

6 Intrusion Detection Alerts

The intrusion detection alerts generated during the investigation pointed to activity consistent with **TrickBot malware**. Several IDS signatures matched patterns commonly associated with TrickBot infections.

The POST traffic also suggested that browser-related information and possible credential data were being transmitted to an external destination. These indicators make it more likely that the traffic reflects a real malware compromise rather than normal user activity.

7 Conclusion

The PCAP analysis showed multiple indicators of malicious activity inside the captured traffic. A Windows executable file was downloaded to an internal host, which suggests the initial stage of infection.

After that, the infected system communicated with an outside server and sent system information through HTTP POST requests. IDS alerts further linked the behavior to TrickBot-related activity, suggesting that the host may have been involved in a broader malware campaign.

Overall, this investigation shows how packet capture analysis can reveal malware delivery, external communications, and possible data exfiltration during a security incident.

8 Non-Technical Summary

The network traffic showed that a computer inside the network downloaded a malicious program from an outside source. After the malware was installed, that computer began sending information to an external server.

Security alerts matched the behavior to patterns commonly associated with TrickBot malware. The infected machine appeared to send system details, including process and browser-related information, outside the network. This suggests the system was compromised and that sensitive information may have been exposed.